



Jan C. Brammer

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I have 5 years of experience in developing open source [software for science](#) and wrestling the [complexity demon](#) along the way. I strive to be a generalist, and I'm most happy when I get to write idiomatic, concise, functional(ish), and type-annotated Python.

Experience

Software developer / RWTH Aachen University / 2021-present

I work on open source data-management tools for chemists under the umbrella of the [NFDI4Chem](#) initiative, collaborating with a distributed team of developers and chemists.

Chemotion ELN

An open source electronic lab notebook for chemists. While implementing new features, I conducted user interviews, analyzed and scoped requirements, wrote code, tests, and documentation. I also reviewed pull requests, improved and maintained our CI, and reduced technical debt by increasing test coverage, removing dead code, and refactoring parts of the codebase (the latter two being my favorite).

Technologies

Javascript, React, Ruby (on Rails), PostgreSQL, Docker, GitHub Actions, GitHub Copilot, VSCode

Links

https://github.com/ComPlat/chemotion_ELN

InChI & TUCAN

Open source identifiers for chemical molecules. These identifiers are to molecules what [ISBN](#) is to books. I developed an extensive test suite and CI for [InChI](#), a legacy C library that is fundamental to commercial and academic chemistry. I also contributed significantly to moving InChI's development to GitHub. TUCAN is a prototype that addresses some of InChI's shortcomings for anorganic chemistry.

Technologies

Python (pytest, pydantic, ctypes, networkx), bash, SQLite, Docker, GitHub Actions, GitHub Copilot, VSCode

Links

<https://github.com/IUPAC-InChI/InChI>, <https://github.com/TUCAN-nest/TUCAN>, <https://doi.org/10.1186/s13321-022-00640-5>

Staff scientist / Radboud University Nijmegen / 2017-2021

I worked in research and development at the intersection of behavioral- and neuroscience. My tasks ranged from software development and data science to experiment design, and the publication of scientific articles.

Biofeedback application development

A virtual reality training to help Dutch police officers regulate acute stress. International, interdisciplinary collaboration of scientists, designers, game



developers, and police. I integrated heart and breathing sensor data into the application, ran extensive user tests, and analyzed requirements. I'm no longer on this project, but I keep maintaining the codebase as a personal project.

Technologies

PySide6 (Qt for Python), Redis, Bluetooth

Links

<https://github.com/JanCBrammer/OpenHRV>, <https://doi.org/10.3389/fpsyg.2021.586553>

Biopeaks

A graphical user interface for the interactive analysis of physiological sensor data. Our lab needed a tool to inspect, clean, and extract features from physiological data.

Technologies

PySide6 (Qt for Python), Python (numpy, scipy, pandas, matplotlib, pytest)

Links

<https://github.com/JanCBrammer/biopeaks>, <https://doi.org/10.21105/joss.02621>

Skills

- (open source) software development
- writing (e.g., technical documentation, scientific articles)
- conducting scientific studies (e.g., experiment design, data- acquisition and analysis)
- data science (e.g., wrangling, visualization, inferential statistics, basic predictive modelling)
- physiological sensor data (e.g., electrocardiogram, photoplethysmography, breathing)
- basic [chemical informatics](#)

Education

MSc Cognitive Neuroscience, Maastricht University, 2015-2017

BSc Psychology, Maastricht University & Concordia University Montreal, 2012-2015